BOSTON
ARCHITECTURAL
COLLEGE
SINCE 1889

DME2100 - Hacking Geometry

SYLLABUS

Instructor: Hamze Machmouchi
Year and term: FALL 2026
Meeting time: Wednesday 4 pm EST
Location: Online via zoom
Credits: 3 credits

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Office Hours: 4PM EST – 6PM EST

1. COURSE INFORMATION

COURSE DESCRIPTION:
This course explores the relationship between architectural geometry and 3D modeling. Through a combination of hands-on physical experiments, theoretical readings, and digital modeling techniques, students will develop a strong understanding of form-finding and the geometric principles behind complex architectural forms. We will investigate topics such as NURBS, SUBD, and mesh modeling, and learn how to create 3D models with an awareness of architectural fabrication processes. This course aims to provide students with a deeper understanding of the relationship between geometric design and digital fabrication, empowering them to not only master the techniques of 3D modeling in Rhino 8 and Autodesk Maya, but also to understand the underlying principles. Students should have basic skills in Rhino and be prepared to learn new software.

This semester the course adds a further step. Students will identify the geometric rule governing their assigned concept and then encode that rule as a working script, using AI tools as a coding assistant. The result is not a single model but a parametric system capable of generating a family of related forms. No prior programming experience is required.

LEARNING GOALS

1. Develop a strong understanding of architectural geometry principles.
2. Gain proficiency in 3D modeling techniques using Rhino 8 and Maya.
3. Apply theoretical knowledge to create informed and innovative 3D models.
4. Bridge the gap between physical experimentation and digital design.
5. Encode geometric rules as functioning parametric tools.
6. Critically analyze and appreciate the role of geometry in architectural design.

COURSE STRUCTURE
This course meets every Wednesday at 4 pm EST via zoom. The course will feature a blend of lectures, hands-on workshops, and presentations by invited experts in the field of architectural geometry. The semester consists of thirteen sessions. Eleven meet on zoom. Two are studio sessions held on Discord, where the instructor is available throughout the scheduled class time.

August 26 - First day of classes
October 14 - Midterm presentations
December 2 - Final presentations and last day of classes

November 11 - BAC closed, Veterans Day, no class
November 25 - BAC closed, Thanksgiving break, no class

WEEKLY SCHEDULE

August 26	First day of classes. Introduction to the course. Students are assigned their individual semester-long concepts, their initial group tasks, and their crit partners. The session closes with a demonstration in which a parametric script is written from scratch, including its errors and revisions.
September 2	Physical Experiments Workshop. Studio session on Discord, no zoom meeting. Groups carry out their form-finding experiments during the scheduled class time. Photographs of at least three experiments, together with a written statement of the geometric rule observed, are due Sunday September 6.
September 9	Concept and Taxonomy. Review of the physical experiments, followed by student presentations of research on the history and geometric principles of their concepts. Each presentation concludes with the student's parameter list.
September 16	Digital Translation and Topology. A workshop on translating physical forms into digital models, including a session on clean mesh topology. The second half of the session is the first scripting lab.
September 23	Surface Analysis and Rationalization. A lecture on understanding surface quality through synclastic and anticlastic curvature, using the Curvature Graph tool in Rhino, followed by a discussion of rationalization for fabrication.
September 30	Parametric Tool Development. Moving from a script that produces a single instance to one that produces a family of forms. The second half of the session is an open debugging clinic.
October 7	Studio Session and Desk Crits. An in-class studio session to develop parametric design rules with instructor support, combined with individual desk crits. The final working session before the midterm review.
October 14	**MIDTERM PRESENTATIONS.** Students present their full process to date: concept, physical experiments, and the resulting digital 3D models. Each student then demonstrates their parametric tool in operation, with a classmate adjusting the parameters during the demonstration.
October 21	Tool Exchange. Studio session on Discord, no zoom meeting. Students exchange parametric tools with their assigned partner and apply the received tool to their own geometry. One image produced with the partner's tool, together with written notes on the difficulties encountered, is due Sunday October 25.

October 28	Exchange Review and Advanced Operations. Discussion of the failures encountered during the exchange and revision of the scripts accordingly. Students are then assigned their Exploration Type.
November 4	Desk Crits. Individual reviews focused on the spatial and architectural qualities of the evolved digital forms. The withdrawal deadline for this course is Friday November 6.
November 18	Visualization Workshop. An introduction to advanced rendering techniques in D5, Twinmotion, Rhino Render, Nano Banana, and GPT Image 2, including methods for scripting repetitive output tasks. The final zoom session before the final review.
November 29	Draft final boards are placed on the studio board by Sunday night. Response partners comment by Tuesday night, December 1.
December 2	**FINAL PRESENTATIONS.** Students present their fully evolved final designs, covering concept, physical experiments, the parametric tool, the evolved form, and final visualizations. Last day of classes.

COURSE EXPECTATIONS

Students are expected to be actively engaged throughout the course. Because the course meets online and only once per week, the following routines are used to maintain continuity between sessions. Each is brief, and each contributes to the participation portion of the final grade.

1. **Weekly pin-up.** Each Sunday night, students will place one image of work in progress in their own frame on the studio board, along with one sentence describing the current difficulty. Work is not expected to be resolved or presentable at this stage. Each frame accumulates over the semester, so that by December it holds a complete visual record of the project.
2. **Response partner.** Each week students are assigned one classmate by name and will respond to that classmate's work by Tuesday night. Responses are left as comments anchored to a specific point on the image, not as general remarks. A response that could have been written without looking at the image does not meet the requirement. Assignments rotate weekly, so that over the semester each student engages with every project in the class.
3. **Crit partner.** Students are paired for the full semester. Crit partners review one another's material before each formal review and serve as a first point of contact for technical difficulties outside class hours.
4. **Prompt log.** Students will keep a record of their exchanges with AI tools while developing their scripts, including unsuccessful attempts and revisions. The log documents the reasoning behind a tool as the sketchbook documents the reasoning behind a drawing. It is submitted at both reviews.
5. **Failures archive.** Students are encouraged to post scripts and models that failed, including parameters driven past their useful range and errors that produced unexpected results. Unintended outcomes are a legitimate source of formal material in this course.

WHERE WORK IS POSTED

Work is posted on the studio board. Conversation happens on Discord. A single FigJam board runs for the whole semester, and each student has a named frame on it which functions as their wall in a physical studio. Weekly pin-ups, tool exchange results, draft boards, and the midterm and final presentations all take place on this board. Critique is written as a comment anchored to a specific point on the work, so that every remark stays attached to what it refers to and remains readable months later. Students should be signed in with their BAC accounts so that comments are attributed.

Discord is used for everything that is not the work itself: announcements and weekly partner assignments, technical help with software and scripts, the failures archive, and a voice channel open during office hours and the two studio sessions. Students are expected to check it at least twice per week.

A shared drive dedicated to this class will be used to share readings, resources, and materials related to coursework. Students are expected and encouraged to upload their progress work to this drive throughout the semester.

FINAL PROJECT
The final project will challenge students to synthesize their knowledge of architectural geometry and 3D modeling skills to design and model a complex architectural form. The project should demonstrate an understanding of form-finding, and the relationship between physical and digital design processes. Students retain the same assigned concept from August through December and develop it across five phases.

PROJECT CRITERIA:

- **Physical Experiments:** Students will conduct physical experiments to explore form-finding and geometric principles. Each student concludes this phase with a written statement of the geometric rule observed in their material, expressed in plain language.
- **Concept and Taxonomy:** Students will select a concept or theme related to architectural geometry, such as minimal surfaces, vaults, or parametric design, and create its geometric taxonomy. Each taxonomy concludes with a list of the parameters that vary within the concept and those that remain fixed.
- **Parametric Tool:** Students will translate the rule identified in the first phase into a working script that generates their geometry from a small set of inputs. Work takes place in the Python editor built into Rhino 8. Students must be able to explain the operation of their own tool, in their own words, at both reviews.
- **3D Modeling:** Students will create accurate and detailed freeform 3D models in Rhino 8 and Maya, demonstrating an understanding of the taxonomy studied in the first half of the semester. Each student is then assigned an Exploration Type and applies it to their system as an additional operation within the existing tool.
- **Visualization:** Students will document their process and present their final project in a clear and compelling way, including rendered images, diagrams illustrating the geometric properties of the concept, and a discussion of the architectural implications of the resulting form.

ATTENDANCE
Students are expected to be present during class. Absences accruing more than three times during the semester will result in a reduction of grades at the discretion of the instructor. If there are any extenuating circumstances, please reach out to the instructor to inform them of your situation and to schedule extensions if permitted. Lateness to class of more than 15 minutes will be counted as an absence unless otherwise communicated to the instructor in advance.

EVALUATION CRITERIA
Assignments: 25%
Attendance and Participation: 25%
Mid Review: 25%
Final Review: 25%
Total of 100%

ASSISTANCE:
Tutorials will be provided throughout the course in addition to technical support on Discord. For all hardware assistance, please contact the Help Desk at help@the-bac.edu or (617) 585-0191.

BAC GRADE DEFINITIONS
The BAC's Grade Definition Chart is included in this syllabus. Students should note that minimum GPAs, determined by their program, are required to maintain Satisfactory Educational Progress. Failure to maintain SEP may result in assignment of additional work, repeating a course or semester, or withdrawal from the program.
More information, including the GPA required for Satisfactory Educational Progress by program can be found at Satisfactory Educational Progress | The BAC (the-bac.edu)

BOSTON ARCHITECTURAL COLLEGE
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GRADE	4.0 SCALE	0 – 100 SCALE	DEFINITION
A	4.0	94 – 100	Excellent. The work exceeds the requirements of the course and demonstrates complete understanding of course goals. In addition, assignments exhibit a level of critical thinking that has allowed the student to demonstrate creative problem solving. Ideas and solutions are communicated clearly, showing a high level of attention and care.
A–	3.7	90 – 93	
B+	3.3	87 – 89	Good. The work meets the requirements of the course and demonstrates understanding of course goals. The assignments reflect an ability to solve problems creatively, but solutions demonstrate inconsistent depth and critical thinking ability. Ideas and solutions are communicated effectively, but may lack the clarity and depth one sees in excellent work.
B	3.0	84 – 86	
B–	2.7	80 – 83	
C+	2.3	77 – 79	Satisfactory. The work meets the minimum requirements of the course and reflects understanding of some course goals but is lackluster. The assignments exhibit a basic problem-solving ability, but the process and solutions lack sufficient depth and demonstrate a need for greater critical thinking. Ideas are communicated ineffectively, showing a lack of attention to detail and a decided lack of clarity or depth.
C	2.0	74 – 76	
C–	1.7	70 – 73	
D	1.0	60 – 69	Less than satisfactory. The work barely meets the minimum requirements of the class. Assignments lack depth and display a minimal understanding of course goals. Ideas are presented with little or no detail or elaboration. Course guidelines are often not followed.
RF Repeat/Fail	0.0	0 – 59	Unacceptable or missing work. The work neither satisfies the requirements of the class nor demonstrates understanding of course objectives. The presentation of work is unprofessional and/or incomplete. Overall, the student shows insufficient understanding of the course requirements. Poor attendance or violation of academic integrity policy may also be factors.
NF	0.0	N/A	
I	N/A	N/A	Incomplete
W	N/A	N/A	Withdrawn
P	N/A	N/A	Pass. Used only in specially designated courses and educational reviews.
NP	N/A	N/A	No Pass
NC	N/A	N/A	No credit. Used if a student replaces a failing grade. Not included in GPA calculation.
NS	N/A	N/A	No show. Awarded only for Educational Reviews if a student registers but does not attend.
T	N/A	N/A	Transfer credit
WV	N/A	N/A	Waiver

2. COURSE POLICIES AND PROCEDURES

DEADLINES POLICY:
Students should complete assignments to the best of their ability and submit them on time. If circumstances require a late submission, the student should contact the instructor before the assignment is due; instructors may consider appropriate accommodation at their discretion. In the event of an emergency (e.g., medical, personal), the student should contact both the instructor and student advisor as soon as possible.

MID-SEMESTER WARNING
Students will receive a progress assessment at mid-semester. To help students succeed, students who do not perform up to expectations will receive a Mid-Semester Warning; a copy of the warning will be kept in the student's file.

WRITING STANDARDS
Writing in this course should meet the standard of accuracy and the clarity of expression expected of design professionals, such as appropriate grammar, correct spelling, and clear and well-organized arguments.

USE OF A.I.
Artificial intelligence tools may be used to assist in developing the scripts and parametric tools required in this course. Their use is treated as a normal part of the working method rather than as an exception to it, and students are not required to avoid them.

What is assessed is not the code but the student's understanding of it. Students must be able to explain what their tool does, how they arrived at it, and the logic on which it operates. A script that produces a correct result but that its author cannot account for does not meet the requirements of the assignment, and will not receive credit on the basis of its output alone.

This understanding is demonstrated in two ways. The prompt log records the development of the tool, including unsuccessful attempts and the revisions that followed them. A written explanation of each function, in the student's own words, is submitted alongside the script at both reviews. Students should expect to be asked to explain any part of their tool during a review.

Work submitted in this course shall be the original creation of its author, and is allowed to contain the work of others so long as there is appropriate attribution. Students who use AI tools for purposes other than script development, including image generation and written research, should indicate where and how those tools were used.

Different courses at the BAC could implement different AI policies. It is the student's responsibility to be aware of and comply with expectations for each course.

The language for our AI policy options draws from the University of Delaware, Duke University, and Harvard College.
Guidry, Kevin. "Considerations for Using and Addressing Advanced Automated Tools in Coursework and Assignments". https://ctal.udel.edu/advanced-automated-tools/. Center for Teaching & Assessment of Learning, November 20, 2024.; "Artificial Intelligence Policies: Guidelines and Considerations." https://learninginnovation.duke.edu/ai-and-teaching-at-duke-2/artificialintelligence-policies-in-syllabi-guidelines-and-considerations/. Learning Innovation & Lifetime Education, August 2024.; "AI Guidance and FAQs". https://oue.fas.harvard.edu/ai-guidance. Harvard College Office of Undergraduate Education, 2025.

3. COLLEGE POLICIES AND PROCEDURES

ACADEMIC INTEGRITY

As stated in the Campus Compact the BAC expects intellectual activities to be conducted with honesty and integrity. Work submitted or presented as part of a BAC course:

- Shall be the original creation of its author;
- Is allowed to contain the work of others so long as there is appropriate attribution; and
- Shall not be the result of unauthorized assistance or collaboration.

Failure to adhere to these guidelines is academic dishonesty and may lead to disciplinary action.

PLAGIARISM

Plagiarism is the act of representing someone else's words, images, or ideas as one's own and is a violation of academic honesty. The institution takes such violations seriously. Please see the policy in the Campus Compact. Any suspected cases of academic dishonesty will be addressed according to this policy.